

Robo-Advisor Adoption, Willingness to Pay, and Trust— Before and at the Outbreak of the COVID-19 Pandemic

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Abstract

We compare readiness to adopt and willingness to pay for (as a proxy for trust) the same financial advice labeled three ways (algorithm, human, and a hybrid of algorithm and human) using an online and a computerized laboratory experiment. Our experiments were conducted before and at the outbreak of the COVID-19 pandemic. We observe a significantly higher readiness to adopt financial advisors at the outbreak of the pandemic, a period of high volatility in the financial markets, compared to before. While in the pre-pandemic period, for the same potential financial advice, we observe age and gender differences in the readiness to adopt and pay for the different advice options, at the outbreak of the pandemic, we observe less sensitivity to the advisor's purported type and no algorithm aversion. Our findings suggest that during a crisis, individuals care more about having an advisor and are less sensitive to the type of advisor, whereas in ordinary times, they are more sensitive to the type of advisor and overall desire less advice. Our findings relate to the financial advisor, trust, artificial intelligence, and information literatures.

Keywords: robo-advisor, willingness to pay, trust, technology adoption, financial literacy, experiment

1. Introduction

This project aims to contribute to our understanding of the factors that affect the potential demand by retail customers for robo-advisors (algorithms), human advisors, and hybrid services (“algo-humans”) before and at the beginning of the coronavirus disease 2019 (COVID-19). Hence, it is related and contributes to the financial advisor literature, the technology adoption literature, and the literature on potential consequences of the recent COVID-19 pandemic.

The academic literature on financial advisors indicates that managing long-term savings is a challenge that involves substantial effort, literacy, time, and sophistication, including setting adequate long- and short-term goals, ensuring sufficient savings amounts, choosing investment assets and instruments, analyzing portfolio yield goals, evaluating one’s risk tolerance, and optimizing one’s asset allocation, level of diversification, and investment horizons, among other tasks. Given the important implications of these decisions and the documented challenges, many investors use investment advisors/portfolio managers to obtain better outcomes.

Certain segments of the population may feel that they lack the knowledge to make financial decisions by themselves or they feel nervous or anxious doing so.^{1,2} To overcome such anxiety, they may establish a “bridge of trust” with a reliable advisor.³ Indeed, the use of financial advisors is widespread. In a market analysis by Mintel (Mintel, 2015), it was reported that 53% of investors in the United Kingdom were assisted by an investment advisor at least once over the previous 3 years. Similarly, in 2012 almost half of Canadians survey respondents

¹ For a discussion about financial literacy in the United States and elsewhere, see Lusardi and Mitchell 2014.

² Certain age groups may be prone to making greater suboptimal financial decisions. For a discussion about how financial mistakes may follow a U-shaped pattern, see Agarwal et al. 2007.

³ As described in Guiso, Sapienza, and Zingales 2004, 2008; Gennaioli, Shleifer, and Vishny 2015; Dorn and Weber 2017; Statman 2018; Peterson et al. 2015.

indicated that they used financial advisors (Foerster et al. 2017, citing a report from the Investment Funds Institute of Canada).

Advances in technologies and algorithms and their use have become common in many aspects of daily life. Hence it is not surprising to see them appear in different segments of the financial industry. In the context of this project, the technology advancements have led to the rise of the “robo-advisor.” Robo-advisors are applications for financial investment planning, based on algorithms that provide advice or invest the client’s assets autonomously, according to the client’s specific financial situation, risk aversion profile, and investment goals.

Although the academic literature exploring whether (and how) robo-advisors offer a substantive advantage over human advisors is relatively preliminary, the adoption of such services is based on the possibility that they can provide added value. These values varied; robo-advisors may generate surplus yields, act with no behavioral biases, and assist investors with better diversification in terms of risk and the number of assets in which they are invested. They may also reduce costs. The fees for a robo-advisor are usually lower than fees charged in the traditional financial advising industry, and robo-advisors are more available to customers since one can use the service with nearly negligible investment amounts, and less than what is usually required by traditional advisors (D’Acunto, Prabhala, and Rossi 2019; Fisch, Turner, and Pension Policy Center 2017; Calvet, Campbell, and Sodini 2007; Foerster et al. 2017; F. Glaser et al. 2019).⁴

The rising use of robo-advisors introduces many potential research questions, some related to the estimation of the abilities of robo-advisors versus human advisors and others to the level of biases that each type may be prone to; our goal, though, is different. We aim to focus and isolate the “trust” component associated with three different advising mechanisms—

⁴ Comparison tables of robo-advisors, including the minimum initial investment, are available from <https://www.nerdwallet.com/blog/investing/best-robo-advisors>.

robo-advisors (algorithms), human advisors, and hybrid services (“algo-humans”)—while controlling for other relevant variables.

The empirical challenge of estimating “trust” differences motivates the use of experimental economics methods. The initial structure of the experiments is based on an investment game that is similar to the game investigated by Mugerman, Sade, and Winter (2021) but with several modifications as follows. The subjects in our experiments chose whether they wanted to use an added-value advising service offered to them and, if so, how much they were willing to pay for it. Unique to our setting, all subjects faced the same information and financial decision with one difference: the label on the type of advice offered to them (algorithm, human, or algo-human). That is, each subject was introduced to one type of advising agent that had the same potential ability as the other two to increase the probability of yielding higher returns (between-subjects design). As the only difference in the manipulations is the “label” of the advisor, a rational agent that operates in an economy without friction should make the same decision regardless of the label. Given the randomization of the subjects to the different treatments, our hypothesis was that we would observe significant adoption of added-value advice and no significant differences in readiness to adopt (RTA) and willingness to pay (WTP) related to the different manipulations. Any observed differences can serve as indications of prior perceptions and preferences related to the given label or type of friction. A willingness to pay a premium for a particular manipulation can indicate greater trust in that kind of potential advice. Our use of the WTP measure as a proxy for trust is consistent with McEvily, Radzevick, and Weber 2012, and Gennaioli, Shleifer, and Vishny 2015, among others. In addition to a full population analysis, we explore gender and age differences, as Venkatesh and colleagues (Venkatesh, Morris, and Ackerman 2000; Venkatesh et al. 2003) and Meyer (2011) found them to be positively and negatively, respectively, correlated with technology adoption.

As trust may be sensitive to financial market conditions, we study what might affect these decisions before and during the high-volatility period in the financial markets brought on by the COVID-19 pandemic. The World Health Organization (WHO) declaration of a worldwide pandemic (WHO, 11.03)⁵ led to an interesting unexpected shock with immediate relevant consequences (immediate high volatility in the financial markets and high reliance on technology).⁶ This raises the question of whether we can observe a change in advice-seeking behavior. The increase in use of and reliance on technology in different areas of one's life because of COVID-19 raises the question of whether this affects the level of trust in and desire to adopt technology-based advisors in particular.

For our investigation, we conducted multiple experiments and employed two experimental methods (online and in the laboratory). Specifically, We ran experiments consisting of a computerized investment decision-making game in one online and one laboratory study before the COVID-19 pandemic (mid-2018, when the stock market trend was up) and two online studies at the outbreak of COVID-19, in a volatile period in the financial markets (in April 2020). See Figure 1 for the experimental design.

[Figure 1]

We conducted the investigation for the total sample and for age and gender subgroups. We also controlled for several variables derived from well-accepted measures in the information and finance literatures that are known to be significant for technology adoption

⁵ Coronavirus Disease (COVID-19) - Events as They Happen by the World Health Organization.

⁶ The pandemic has led many governments to unprecedented acts (e.g., limiting flights and arrivals of foreign citizens and issuing lockdown orders). These acts, which restricted economic activity by closing business and markets and reducing international trade, had immediate economic consequences. Global financial markets reacted with significant drops in one month. For example, on February 19, 2020, the S&P 500 traded at 3,386, but by March 23, it had plunged to 2,237. The European and Israeli stock exchanges behaved in a highly correlated and similar manner. At the same time, these travel restrictions, extended lockdowns, and social distancing conditions put in place in response to the pandemic have led to a profound change in how people use digital technologies to work, learn, consume services, buy products, or even play. Many services, by governments and by the private sector, including financial institutions, have switched to a digital-only format that requires a change in habits.

and savings decision making. For technology adoption, we used the Unified Theory of Acceptance and Use of Technology 2 (UTAUT2) developed by Venkatesh, Thong, and Xu 2012.⁷ To assess financial literacy, we used a measure based on an index developed by Lusardi and Mitchell (Lusardi and Mitchell 2011, 2007). Last, we controlled for quantitative knowledge and demographics.

The results obtained before the COVID-19 pandemic indicate significant age and gender differences in the RTA and WTP measures for the different suggested mechanisms even after controlling for other relevant variables. In Study 1, an online experiment with a representative sample consisting of 612 subjects, we find no statistically significant differences between the advice mechanisms in the RTA or the WTP for the total sample. However, age plays a role in the RTA and WTP for the different advising mechanisms. Specifically, for the second of three age groups (31- to 44-year-olds),⁸ we find significant algorithm aversion. Subjects' RTA and WTP were lower for algorithmic advice. The addition of human assistance to the algorithm (algo-human alternative) significantly increased the subjects' acceptance rate and WTP (algorithm RTA = 0.51 and WTP = 5.97; algo-human RTA = 0.77 and WTP = 14.5; and human RTA = 0.76 and WTP = 13.75). We also document gender differences. We find that men were willing to adopt and pay significantly more for advice in general (RTA = 0.73 and WTP = 11.7 for men and RTA = 0.6 and WTP = 9.51 for women) and that they were ready to adopt and willing to pay significantly more than women for the algorithmic advice (RTA = 0.75 and WTP = 12.35 for men and RTA = 0.57 and WTP = 7.63 for women). Also, men were more ready than women to adopt human advice (RTA = 0.71 for men and 0.61 for women) but not willing to pay more than women for it (WTP = 11.46 for men and 10.61 for women).

⁷ See Section 2 and Appendix C for more details.

⁸ We provide robustness checks for the age-span choice in Appendix E.

The experimental results suggest that the WTP measure provides an additional insight beyond RTA. As “trust” is widely used in the academic literature in different contexts, by using different measures, we investigate the correlation of the WTP for each alternative in our research to some of these accepted measures.⁹ We find positive significant correlations only between WTP and calculational trust and to some degree distrust antecedents. These results suggest that subjects were willing to pay a premium for all advice alternatives when they felt more comfortable with the alternative. Also, when levels of distrust are high, subjects wanted to transfer responsibility and avoid making decisions by themselves and were willing to pay a premium for algorithmic advice.

While conducting experiments via an online provider (compared to a laboratory environment) has advantages such as accessibility and wider representation of different segments of society, it also has disadvantages, such as not being able to observe the subjects as they work on the tasks. Hence, before the COVID-19 pandemic, we also conducted a laboratory experiment with students (a younger, educated age group; Study 2). The use of the laboratory enabled us to closely monitor the process in a more controlled environment.¹⁰

The subjects in Study 2 were students at the Hebrew University, Jerusalem, Israel. Their average age was 22.8 years. Our laboratory results on WTP indicate that subjects who have been offered algorithmic advice are, on average, willing to pay significantly more for the

⁹ See, for example, McKnight, Choudhury, and Kacmar 2002; Gefen, Karahanna, and Straub 2003; Komiak and Benbasat 2006. Specifically, we use “institutional trust” (environmental and not direct trust), “calculational trust” (trust based on the rational explanations from all available information generated), and “anxiety” (distrust) measurements. All of these measures are based on direct questions to subjects. For descriptions and more discussion, see Section 2 and Appendix B.

¹⁰ Our project joins others in the financial economics literature that use several experimental methods in a particular study to enrich the discussion and overcome some of the challenges of each method. See, for example, Glaser, Iliewa, and Weber 2019; Haigh and List 2005; Hurwitz, Sade, and Winter 2020; Camerer et al. 2016; Friedman, Friedman, and Sunder 1994; Bosch-Domenech et al. 2002, among others. Our project also joins the line of financial economics research that uses different types of subjects. For an additional discussion and examples from the literature about the use of different types of subject pools in experiments see, for example, Sade, Schnitzlein, and Zender 2006; Haigh and List 2005; Holzmeister et al. 2020, among others.

algorithmic advice, supporting a nonsignificant tendency toward algorithm advice found in Study 1 (WTP = 13.76, 8.54, and 10.54 for the algorithm, algo-human, and human advice, respectively).¹¹

Study 3 was conducted at the start of the COVID-19 pandemic (mid-April 2020). The results provide additional insight on the demand for financial advice during a unique, unexpected shock to the economy and people's livelihood. We have the ability to compare results obtained before and at the outbreak of the crisis, as we conducted this online experiment with similar characteristics to Study 1 and with the same online provider (with a representative sample of 526 participants) when the pandemic was beginning to affect Israel. See Figure 2 for the results comparison.

[Figure 2]

Interestingly, results suggest that overall, participants were significantly more ready to adopt advice at the outbreak of the COVID-19 crisis compared to before the pandemic (before COVID-19, Study 1, RTA = 0.66; at the outbreak of COVID-19, Study 3, RTA = 0.73).¹² Even though there is an overall higher demand for advice in the experiment conducted at the beginning of the pandemic, we observe less sensitivity to the advisor's purported type and no algorithm aversion for the middle age group (31- to 44-year-olds), as we document in the period before COVID-19. Our results for the experiment conducted at the outbreak of the COVID-19 pandemic indicate that gender differences remain significant; men are willing to adopt and pay more for advice.

¹¹ Age differences found in Study 1 and confirmed in Study 2 might also be generational differences. Given the limited time frame of the experiment, we could not make a distinction between the two plausible explanations.

¹² Additionally, the two student experiments provide similar conclusions (before COVID-19, Study 2, RTA = 0.75; at the outbreak of COVID-19, Study 4, RTA = 0.85), though admittedly due to the differences in the experimental methods this comparison is limited. The student experiment before COVID-19 was conducted in a controlled laboratory environment, whereas COVID-19 restrictions in place during Study 4 meant the laboratory was off-limits and this student experiment was conducted online, with no real monetary consequences and with only limited questionnaires that were linked to the decision-making game.

Our project is relevant to the domains of COVID-19 financial ramifications, financial advice, algorithmic advice (robo-advisors), human–computer interaction, technology adoption, and financial literacy and contributes to the literature in the following ways: First, we have had a unique opportunity to conduct financial-decision experiments regarding financial advisors before and during a major unexpected pandemic event that has had substantial financial and economic consequences. Second we study perceived trust in the form of WTP in the relatively new and growing area of financial robo-advising, comparing algorithmic advice to two possible alternatives, a human advisor and an algo-human hybrid, in different age and gender groups. Third, we contribute to the information and technology literature on the adoption of algorithm-based services in various areas of life. We emphasize the need to estimate the WTP in addition to the RTA (the common measure in the technology-adoption literature). Fourth, we add to the discussion of the advantages and disadvantages of using online versus laboratory experiments.

Next, in Section 2 we provide a relevant literature review and hypotheses, and in Section 3 we provide a detailed description of the methods and results of the two experiments conducted before the COVID-19 pandemic: Study 1 online and Study 2 in the laboratory. In Section 4 we describe the methods and results of Studies 3 and 4, conducted at the outbreak of the COVID-19 pandemic, and in Section 5 we conclude.

2. Related literature and hypotheses

In this section, we review the relevant literature on financial advice, gender and age, robo-advisors and technology adoption, and trust and set our hypotheses.

2.1. Financial advice

Managing savings is a task that many people reluctantly take up (Gennaioli, Shleifer, and Vishny 2015). This is not surprising, as managing long-term savings properly involves many challenges. The financial economics literature deals extensively with the question of

whether investment consultants indeed provide an effective service to their clients and if so, what value they create. The literature discusses three possible benefits (not mutually exclusive) that advisors can generate for their customers: (a) Advisors may generate surplus returns on their clients' investments (in investment terms, "Alpha"; Berk and Green 2004; Campbell 2006; Bergstresser, Chalmers, and Tufano, 2009; Chalmers and Reuter 2012; Hackethal, Haliassos, and Jappelli 2012; Foerster et al. 2017; (b) they can advise on issues such as risk and asset diversification (Calvet, Campbell, and Sodini 2007; Badarinza, Campbell, and Ramadorai 2016; Foerster et al. 2017; D'Acunto, Prabhala, and Rossi 2019); and (c) they may reduce anxiety that some individuals have with respect to financial planning and the financial markets by enabling clients to transfer responsibility to someone whom they trust (Gennaioli, Shleifer, and Vishny 2015; Guiso, Sapienza, and Zingales 2004, 2008).

Given that many studies show that investment advisors create, at least before management fees, a certain Alpha (e.g., (Jensen 1968; Bernstein 1993; Campbell, Viceira, and Viceira 2002; Berk and Green 2004), we set up our experiments such that the various types of consultants that we investigate (algorithm, human, and algo-human) create in expectation a positive Alpha with a lower variance before fees. Gennaioli, Shleifer, and Vishny's (2015) theoretical model relates the fees charged by advisors to the trust they produce and not only to the financial value they generate. Individuals are willing to pay a premium for an investment consultant they trust more.

Our experimental design is adapted from Mugerman, Sade, and Winter (2018), a study of different payment structures for financial advisors. Though their hypothetical game is similar to ours, Mugerman, Sade, and Winter (2018), investigated different research questions, focusing on a hypothetical human advisor in the lab. They find, using a different sample of subjects and in a different time frame from ours, a significant level of desire to adopt the advisor.

Given the above and with an expectation of consistency, our first hypothesis follows:

Hypothesis 1: A significant number of participants will be ready to the adopt human advice offered in all experiments.

In our experiments, we offered the same expected added value for the advising services in all manipulations, meaning all subjects faced the same information and financial decision with one difference: the label on the type of advice offered to them: algorithm, human, or algo-human. As the only difference in the manipulations is the “label” on the advisor, we hypothesize that a rational agent that operates in an economy without frictions should make the same decision regardless of the label. Given the randomization of the subjects to the different treatments, our hypotheses for the total sample are as follows:

Hypothesis 2: A significant number of participants will be ready to adopt advice in all three label manipulations.

Hypothesis 3: The advice label will not affect levels of RTA and WTP in the total sample.

2.2. Trust measurement

There is no single definition of trust in the academic literature; hence, quantifying trust is a challenging task that researchers address with several approaches. One approach measures trust in a game theory framework and mostly through “investment games” (simultaneous or multi-stage). In these games, the trustor gives the trustee the power to make a decision that ultimately affects the amount of money the trustor will receive (e.g., Yaari and Bar-Hillel 1984; Berg, Dickhaut, and McCabe 1995; Konow 2000; Fehr et al. 2003; Cappelen et al. 2007; Sutter and Kocher 2007). Another approach that is widely used in psychology and sociology is questionnaires, which vary from the broad General Social Survey for Personality Trust to more targeted questionnaires that attempt to examine a particular construct of trust or trust in a specific situation (e.g., Rousseau et al. 1998; Gefen, Karahanna, and Straub 2003). This project

utilizes differences in WTP as a measurement of trust that is consistent with McEvily et al. (2012) and Gennaioli et al. (2015).

Trust can arise for different reasons. Our experimental design enables us to study correlations between three relevant types of trust antecedents and the measurement of WTP: (1) *cognitive and emotional trust* (e.g., McKnight, Choudhury, and Kacmar 2002; Gefen, Karahanna, and Straub 2003; Komiak and Benbasat 2006) as part of what is called calculative trust (Rousseau et al., 1998), which refers to trust based on first impressions of the signals that the trustor takes into account to form an opinion that builds trust (e.g., competence, integrity); (2) *anxiety trust (distrust)*, which according to Pavlou and Gefen (2005), deals with a mindset of suspicion and the level of fear of others. Dimoka (2010) finds that distrust triggers brain areas associated with anxiety, loss aversion, and avoidance; and (3) *institutional trust*, which is impersonal and derives from the environment and context that assists in its creation, such as regulations and guarantees from the environment (e.g., McKnight, Cummings, and Chervany 1998; Rousseau et al. 1998; Huang et al. 2003). In addition, and to further isolate the target trust, we control for several variables derived from well-accepted measurements in the information science and finance literatures that have been documented as being significant for technology adoption and savings decision making (Lusardi and Mitchell 2011; Venkatesh, Thong, and Xu 2012).

For technological RTA, in this research we use the UTAUT2 (Venkatesh, Thong, and Xu 2012), a comprehensive measure that is based on more than 13 other acceptance models that influence the individual intention to use and acceptance of new technologies. It is organized along seven main constructs: (a) performance expectancy—the belief in how beneficial the technology will be; (b) effort expectancy—how easy it is to use; (c) social influence—what one’s important others think; (d) facilitating conditions—what resources and support are available; (e) hedonic motivation—degree of pleasure; (f) price value—the cost–benefit trade-off; and (g) experience and habit—acting automatically through learning. For a more elaborated discussion of the model, see Appendix C.

Financial literacy, as stated above, may play an important role in the desire to hire a financial advisor. In our project we use the standard index that consist of three principal questions developed by Lusardi and Mitchell (2007, 2011) as the control. For a more elaborated discussion on financial literacy, see Appendix D.

2.3. Robo-advisors and technology adoption

Robo-advisors are computer applications for financial investment planning based on algorithms that advise or invest the client's assets, autonomously, according to the specific financial situation, the client's risk aversion profile, and his or her investments goals. Since it is possible to deliver algorithmic counseling with little to no human intervention to a large number of customers, and because the infrastructure of robo-advisor companies is evaluated by many as far less expensive to maintain than that needed to support human financial advisors, the documented fees charged to clients advised by algorithmic advising platforms are usually lower compared to the fees charged by the traditional financial advising industry (Fisch, Labouré, and Turner 2019; D'Acunto, Prabhala, and Rossi 2019). Furthermore, unlike traditional financial advisors who generally limit their services to relatively wealthy customers with a minimum managed asset of USD 100,000, robo-advisor services are available to customers with nearly negligible initial investment amounts (Fisch, Labouré, and Turner 2019; D'Acunto, Prabhala, and Rossi 2019).

Although the research on robo-advisors is still relatively limited, the academic literature investigating technology acceptance in different areas of life is extensive. Theories and models have been developed in behavioral decision making in psychology, information management, and organizational management. We relate our work to the technology adoption literature findings indicating that different segments of the population may be prone to different paces of technology adoption. In particular, gender and age were found to be significant factors.

Following Venkatesh, Morris, and Ackerman (2000), who documented that men demonstrate a greater tendency to adopt technology, and Meyer (2011) and Venkatesh et al. (2003), who documented that younger people are more likely and faster to adopt technology-based solutions, we conjecture that we may observe subgroup differences, indicating existing prior perceptions and preferences, in the RTA and WTP depending on the advice label given:

Hypothesis 4: RTA and WTP for the advice alternative labeled algorithmic will be negatively correlated with age.

Hypothesis 5: RTA and WTP for the advice alternative labeled algorithmic will be positively correlated with gender (male).

2.4. COVID-19

The outbreak of COVID-19 had immediate implications on the financial markets by increasing the uncertainty and volatility in the global financial markets (e.g., Choi 2020; Altig et al. 2020). The high levels of ambiguity during the outbreak may lead to differences in advice seeking compared to pre-COVID times. Though there are no actual differences in the game that we offer to the subjects before and at the beginning of the COVID-19 outbreak, individual emotions and state of mind may have an effect. Hence,

Hypothesis 6: The COVID-19 outbreak will have an impact on the RTA level compared to pre-COVID-19 times.

The COVID-19 pandemic has led to travel restrictions, extended lockdowns, and social distancing mandates that have had a profound effect on how people use digital technologies to work, learn, consume services, buy products, or even play. Many services, offered by governments and by the private sector, including financial institutions, have been switched to a digital-only format that requires a change in habits (e.g., Zeng, Chen, and Lew 2020; Teles, Sacchetta, and Matsumoto 2020). The increase in use and reliance on technology in different

areas of one's life may increase the willingness to adopt technology-based financial advice. Hence,

Hypothesis 7: The COVID-19 pandemic will positively correlate with RTA and WTP for technology-based advisors.

3. Before COVID-19

3.1. Study 1: Before COVID-19—Online experiment with a representative sample

3.1.1. Method

Study 1 was conducted mid-2018 (June–August), before the outbreak of the COVID-19 pandemic, and employed a between-subjects design with three main stages. First, the subjects completed a quantitative effort test. Stage 2 was the investment game that would allow us to elicit trust with respect to the different advising platforms. Stage 3 consisted of a series of questionnaires on technology adoption, financial literacy, and personal characteristics.

The investment game included three rounds. Subjects had to choose whether they wanted to use an investment advising service (the treatment) and how much they were willing to pay for it. Subjects were assigned to groups that were each offered one type of labeled investment advisor: a robo-advisor (hereafter, algorithm), a human advisor, or an algo-human advisor. The label of the advisor was the only difference in the three manipulations.

The experiment was conducted as a computerized survey that could be completed in 30 min or less through Panels4All.¹³ There were 827 initial respondents. As concerns have been raised in the academic community about the use of web-based task providers such as Panels4All or Amazon's M-Turk (because of the potential for responses being from bots or

¹³ Panels4all was established in 2006. According to the company, the panel includes tens of thousands of members of a representative population who fill out surveys and perform experiments. The panel maintains the privacy of the respondents.

other unidentified sources and the lack of true effort and attention), we adopted two exclusion criteria: (a) We required a minimum grade on a five-question quantitative effort test of 20% (62 respondents excluded). (b) We set the minimum length of time for completing the experiment to 10 min (an additional 153 subjects who completed the experiment in less time excluded). Our final sample thus consisted of 612 subjects (mean age = 37.6 years; 48.8% male), with 190 randomly assigned to the algorithm, 203 to the algo-human, and 219 to the human advisor treatment. Table 1 shows the total sample's descriptive statistics and the descriptive statistics of each of the subsamples that were randomly assigned to the three manipulations.

[Table 1]

At the beginning of the experiment, subjects received guided online instructions. After the third stage, we also asked several questions that allowed us to evaluate the extent of the subjects' understanding of the procedures. Most of the subjects stated that the experiment questions were understood.

All monetary values were denominated in an experimental currency referred to as Zuzim (Zu).¹⁴ Explanation of the experimental currency was provided at the beginning of the experiment; the subjects' payments were dependent on their results and decision making in the experiment.

3.1.2. Design

In Stage 1, all subjects were given part of the quantitative section of the Psychometric Entrance Test, a tool used to predict academic performance, which requires five questions to be answered within 5 min. Subjects who passed the test were those who successfully answered at least one question. They received Zu 40 and continued to the decision-making game (Stage 2). Those who did not pass the test received an additional Zu 12 and did not continue the

¹⁴ The exchange rate was 0.25 new Israeli shekel (NIS) for each Zu.

experiment. Subjects did not know in advance how many correct answers were required to pass the test. The purpose of this stage was twofold: First, we wanted the subjects to treat their experimental account in the same manner as they would treat money they earned while investing effort (theory of mental accounting; e.g., Henderson and Peterson 1992; Thaler and Johnson 1990; Corgnet et al. 2015, among others). Second, this quantitative ability measure serves as an additional variable in the characteristics of the subjects, and we can use it as an additional control.

The decision-making game (Stage 2) was designed to test the RTA and WTP for investment advice. The subjects were assigned randomly to one of the three manipulations (algorithm, algo-human, human). Note, the underlying mechanism of the decision game was exactly the same for the three manipulations. Each subject participated in only one type of proposed consultation. The game (based on Mugerman, Sade, and Winter 2018) consisted of three decision rounds. In each round, the subjects were exposed to the same tree of probabilities for winning financial assets. Consistent with other experimental economics and finance work, we used three rounds to verify subjects' learning and understanding of the game. To encourage the subjects to treat each round seriously and separately, they were told in advance that their payment would be based on the results obtained in only one randomly chosen round. Following the common practice in the experimental economics literature of treating first rounds in multi-round experiments as learning trials and analyzing only later rounds (e.g., Dufwenberg, Lindqvist, and Moore 2005; Hussam, Porter, and Smith 2008; Huber, Kirchler, and Stöckl 2016, among many others), we present the results of the third round. To provide evidence that a specific round choice does not determine the conclusions, we also provide the results of Round 2 in Appendix E.

In the investment game, the subjects were presented with the following asset probabilities (Figure 3A): a 40% chance of gaining Asset A, with a probability of 80% of

yielding Zu 40 and 20% of winning 0; and a 60% chance of gaining Asset B, with a 20% probability of yielding Zu 40 and 80% of winning 0. Subsequently, the subjects were offered an investment “advising service” that could improve their probability of making a profit by increasing the probability of obtaining Asset A from 40% to 90%. The structure is shown in Figure 3B.

[Figure 3]

After evaluating the advising service option, subjects indicated whether they wished to use the service and if so, what amount they were willing to pay for it. The subjects were not exposed to the cost of the service, which was determined in a random manner (uniform distribution). Yet, they were told that if the amount they offered to obtain the service was greater than or equal to its cost, the agent would provide the service in return for its cost; otherwise, no service would be provided (Appendix A).

Following the academic literature documenting that advisors can create positive Alpha (before advising fees), we constructed the experiment so that the advice provided could yield value, on average (before advising fees). This value was the same across the advice alternatives. The expectation in all of them, for a risk-neutral agent, was Zu 12 (the expected value without the advisor was Zu 17.6 and with the advisor, Zu 29.6). During the first round, after the investment decision and before the subjects obtained their results, they were required to answer a trust questionnaire (based on McKnight, Choudhury, and Kacmar 2002; Mugerman, Sade, and Winter 2018; Komiak and Benbasat 2006) about the proposed advising service.

In the third stage of the experiment, subjects completed a personal questionnaire with demographic and behavioral details, a questionnaire on financial literacy and knowledge (based on Lusardi and Mitchell 2011), and a questionnaire on technological RTA based on the UTAUT2 measure (Venkatesh, Thong, and Xu 2012). For the detailed questionnaires, see Appendixes B–D.

3.1.3. Results

We compare the WTP and RTA of the different manipulations and further use several relevant control characteristics to explore RTA and WTP: gender, age, trust antecedents, quantitative knowledge, financial literacy, and attitude toward technology adoption. We start with the RTA analysis and follow with the investigation of the WTP. Results reported are from the third round of the experiment.¹⁵

3.1.3.1. Differences in RTA

First, we present the descriptive statistics and then we estimate a multivariate model while controlling for technology adoption, financial literacy, quantitative knowledge, and demographics. We start with the analysis of the total sample and proceed with age and gender subsamples (Figure 4).

[Figure 4]

For the total sample, we find no significant differences between the three advising alternatives. The mean RTA is 0.67, 0.67, and 0.66 for the algorithm, algo-human, and human alternatives, respectively, supporting Hypotheses 1, 2 and 3.

Next, we divide the sample into three age groups (20–30 years, 31–44 years, and 45+ years) that potentially differ in their financial status.¹⁶ For the 20- to 30-year-olds, we find no significant differences in the RTA for the three advice alternatives (RTA = 0.63, 0.66, and 0.59 for the algorithm, algo-human, and human alternatives, respectively). For the 31- to 44-year-olds, we find that subjects were ready to adopt the human advisor (RTA = 0.76) significantly more (by 50%) than the algorithm (RTA = 0.51). Adding human assistance to the algorithm (algo-human) increased the subjects' acceptance in comparison to the algorithm alone, but their RTA for the algo-human (0.77) is not significantly different from the RTA for the human

¹⁵ We analyze the decision making on Round 2 as well (see Appendix E). The main results hold.

¹⁶ There is no common practice in the finance literature for dividing samples into age groups. In the robustness tests (Appendix E), we provide additional analysis based on four instead of three age groups (20–30, 31–40, 41–50, 51+ years).

advice (0.76). For the 45+ group, we find a nonsignificant preference for algorithmic advice (RTA = 0.77) over the other alternatives (RTA = 0.69 and 0.66 for the human and algo-human advice, respectively). Our results are only partially consistent with Hypothesis 4. We find a U-shaped adoption of algorithmic advice and algorithm aversion (e.g., Dietvorst, Simmons, and Massey 2015) for the middle age group (that were mitigated by the hybrid option). The decline in adoption was not documented for the oldest age group of 45+.

When we explore gender differences, we find that men were significantly more likely than women to use advice in general (RTA = 0.73 for men and 0.60 for women). The significant differences are driven by the algorithm and human alternatives: We find that men were significantly more willing to adopt these alternatives than women (RTA = 0.75 and 0.71, respectively, for men and only 0.57 and 0.61, respectively, for women) supporting our Hypothesis 5.

Given that the academic literature (e.g., Morris, Venkatesh, and Ackerman 2005; Lusardi, Mitchell, and Curto 2010; Carlin, Olafsson, and Pagel 2017) indicates age and gender might be correlated with other related variables, such as financial literacy, technology adoption, quantitative knowledge, and demographic variables, we control for these variables. In the first set of regressions, we examine the degree of RTA for the different advice alternatives for the total sample and the different age groups. The base specification is

$$(1) \text{RTA}_{Age,i} = \alpha + \beta T_i + \delta FL_i + \gamma X_i + \varepsilon_i$$

where the outcome variable is the RTA for individual i in the different age groups, vector T is a dummy variable for the type of investment advice: algorithm or algo-human (where the base is the human advice), FL is a financial literacy index indicator (based on Lusardi and Mitchell 2011; see Appendix D for the detailed questions), and the vector control variables X_i consist

of quantitative knowledge (the number of right answers on the psychometric test) and dummy variables for gender (male = 1), religiousness (perceives self as religious to some extent), and work (full- or part-time employment). Table 2 shows the probit analysis results.¹⁷

[Table 2]

From the regression tests on the total sample and the different age groups, we find the same main results as described above. The 31- to 44-year-olds were ready to adopt the algorithmic advice significantly less than the human alternative. The other two age groups are not significantly different from each other in their RTA. In addition, quantitative knowledge is significantly positively related to the RTA in most cases (total sample and age groups 20- to 30-year-olds and 31- to 44-year-olds). The last result is consistent with the expectation that quantitative ability is correlated with better understanding of the game.

We next estimate the following regression model to examine the gender differences in the RTA of the different alternatives. We examine the differences for each of the advice alternatives.

$$(2) \text{RTA}_{\text{Treatment},i} = \alpha + \beta_i \text{Gender}_i + \beta_i \text{Technology}_i + \delta \text{FL}_i + \gamma X_i + \varepsilon_i$$

where the outcome variable is the RTA for each treatment (algorithm, algo-human, human) and individual i is a function of Gender_i (a dummy variable where male = 1), and Technology symbolizes the six constructs of the UTAUT2 (see Appendix C for the detailed questions). Table 3 shows the probit analysis results. The multivariate estimation results confirm the previous results with respect to gender. Men tended to adopt significantly more advice than women in general. In addition, men were ready to adopt the algorithm alternative significantly more than women. The gender difference is significant even though we control for financial literacy and quantitative ability—variables that can be correlated with gender.

¹⁷ We repeated the same analysis using other estimation models, that is, logit and ordinary least squares (OLS), including robust standard errors. The conclusions are not affected by the type of estimation. These were performed for all the regressions in the study and did not affect the main results.

[Table 3]

One could argue that the tendency to adopt technology in general is an important potential explanation with respect to the RTA for the different advising possibilities and in addition, gender may be correlated with technological RTA. When we examine the correlation between RTA and technology adoption characteristics according to the UTAUT2 framework (see Table 3), we find the same main results as reported above. Overall, men were ready to adopt more advice, and we find that this was driven mainly by the algorithm label alternative. We find that individuals with higher quantitative knowledge are ready to adopt significantly more algorithmic advice and marginally more algorithmic advice. For the technology adoption constructs (UTAUT2 framework), we find that the participants who perceived technology as useful (performance expectancy) were marginally more ready to adopt algorithmic advice. Surprisingly, individuals were ready to adopt algorithmic advice significantly less when they perceived their resources and support for adopting advice were high (facilitating conditions). These results are only partially consistent with the literature (Davis 1989; Gefen, Karahanna, and Straub 2003; Venkatesh, Thong, and Xu 2012; Venkatesh et al. 2003).

3.1.3.2. Differences in WTP

Given the way our experiment is structured, the only difference in the three manipulations is what we label each advisor; hence a systematic difference in the subjects' WTP can serve as a measurement of trust differences for each of the three advisor alternatives. Estimating the WTP enriches the discussion regarding the different alternatives, as one might be willing to adopt each of the mechanisms, yet at a different price. The analysis continues with the same methodology performed with the RTA. The analysis for the WTP is performed so that the willingness value for adopters is their stated number and for nonadopters the value is set at zero.

To distinguish any trust effect from other possible explanations, we control for the same variables described in the RTA section: financial literacy index indicator (based on Lusardi and Mitchell 2011), quantitative knowledge, dummy variables for gender (male = 1), religiousness, work, and technology adoption constructs, and now we add trust antecedents as well (see Appendix B for the detailed questions). Figure 4 shows the means of the WTP for the different advice alternatives grouped by age and gender. For the regression analysis, the dependent variable is WTP and we use tobit regressions.¹⁸

Supporting Hypothesis 3, we find no significant differences in the mean WTP between the advising alternatives for the total sample (WTP = 10.02, 10.64, and 11.01 for the algorithmic, algo-human, and human alternatives, respectively); we do observe differences by age and gender.

For the 31- to 44-year-olds, we find that subjects were willing to pay significantly more for the human advice (WTP = 13.75) than for the algorithmic advice (WTP = 5.97). Adding human assistance to the algorithm (algo-human) changed subjects' preferences positively (WTP = 14.5). The results are consistent with the conclusions obtained from the RTA analysis and only partially support Hypothesis 4, as we find a U shape for the WTP for algorithmic advice. For the youngest age group, we find a WTP for alternatives that include algorithmic advice (WTP = 10.8 for the algorithm and 11.03 for the algo-human). Neither is significantly higher than for the human advice alternative (WTP = 9.1).

For the 45+ group, we find that subjects were less willing to pay for the algo-human alternative (WTP = 8.6) compared to the human advice (WTP = 12.45). The WTP for this alternative (algo-human) is not significantly inferior to the WTP for the algorithmic advice

¹⁸ Hence, the specifications are as in Equations (1) and (2) with $WTP_{Age,i}$ as the dependent variable. Furthermore, we add the trust antecedents as an additional set of controls in a separate estimation as $WTP_{Treatment,i} = \alpha + \beta_i Gender_i + \beta_i Trust_i + \delta FL_i + \gamma X_i + \varepsilon_i$. The results are provided in Table 2 columns 5–8 and Table 3 columns 5–11.

(WTP = 11.07). We find no significant differences between the algorithm and human alternatives. That is, although this age group exhibits a nonsignificant lower RTA for the algo-human alternative compared to the human alternative, they were willing to pay significantly less for the algo-human choice.

We find that men were willing to pay significantly more for advice in general (WTP = 11.7 for men and 9.5 for women) and that they were willing to pay significantly more than women for the algorithmic advice (WTP = 12.35 for men and 7.62 for women). This is consistent with our RTA results and supports Hypothesis 5.

When we examine the correlation between the WTP and the technology adoption constructs of the UTAUT2 framework, we find that those who perceived technology as effective (performance expectancy) were willing to pay significantly more for the algorithmic advice compared to the human alternative. In addition, and in keeping with the results on the RTA, subjects who perceived that they had the means and facilitating conditions (i.e., resources and support) were willing to pay less for the algorithm-alone alternative. Furthermore, subjects who perceived technology as easy to use (effort expectancy) were willing to pay marginally less for the algo-human advice alternative. These results are partially consistent with the literature (Davis 1989; Gefen, Karahanna, and Straub 2003; Venkatesh, Thong, and Xu 2012; Venkatesh et al. 2003).

As described above, we believe that the WTP premium for particular advice alternatives derives from trust differences that affect the subjects' perception of the alternatives presented (see Appendix B for descriptions of trust antecedents). We find support (Table 3, columns 9–11) for our expectations in subjects' willingness to pay significantly more for all advice alternatives when they felt more comfortable with the alternative (calculational/emotional trust). In addition, we find that when levels of anxiety were high (distrust), subjects were willing to pay a premium for algorithmic advice when they wanted to transfer responsibility

and avoid making decisions by themselves. Last, for the human advice alternative, we find that the more the subjects assessed the regulatory environment as supportive and strong (institutional trust), the less they were willing to pay for the human advice.

We have conducted several robustness tests for validating the main results of the experiment. We add an analysis with four age groups (20s, 30s, 40s, 50s and above). We find that 31- to 40-year-olds are less ready to adopt and less willing to pay for the algorithm alternative, and for the 41- to 50-year-olds, we see that there is less of a WTP for alternatives that include an algorithm compared to purely human advice. Also, we examine the decision making on Round 2. The main results hold. Furthermore, we add an analysis with all control variables together to ensure the results hold and find they do. For a further elaboration of the robustness tests, see Appendix E.

3.2. Study 2: Before COVID 19—Laboratory experiment with a student sample

In Study 1, we see a certain tendency, although not significant, for the younger age group (20- to 30-year-olds) to pay more for the algorithmic alternatives than the human-alone alternative. In Study 2, we expand the experiment by focusing on this age group in a laboratory experiment. In this experiment, 20- to 30-year-old subjects were randomly assigned to one of the same three advice alternatives we used in Study 1 and played the same investment game.

3.2.1. Method

As in Study 1, we used a computerized experiment, but this time in a laboratory at the Hebrew University. This enabled us to better observe subjects' investment choices and level of focus in the various controlled groups. As in Study 1, the experiment consisted of three stages: The first consisted of three questionnaires, the second was the quantitative effort test, and the third was the investment decision game. The decision-making game was carried out in one round with subjects facing the same decision structure as described for Study 1 (with exceptions noted in the following Design section).

In the laboratory, each subject was placed at a separate workstation so that they would not influence each other's decisions. The sample comprised undergraduate and graduate students from the Hebrew University. Table 4 shows the general descriptive statistics. Of the total sample of 153, 51 subjects were randomly assigned to the algorithmic advice treatment, 52 to the algo-human alternative, and 50 to the human advisor. The average age of the subjects was 22.8 years, and 37.6% were men. Approximately 96% of the subjects were undergraduate students, 50.7% had a full- or part-time job, and 85% were born in Israel, but all spoke Hebrew. The average WTP was NIS 10.86. At the beginning of the experiment, subjects received guided instructions. After the third stage, a summary questionnaire was conducted to evaluate the extent of the subjects' understanding of the procedures. At the end of the experiment, subjects received a payment ranging from NIS 30 to NIS 80 according to their experimental results.

[Table 4]

3.2.2. Design

The procedures differed from those in Study 1 in the following ways: The three questionnaires on personal characteristics, financial literacy, and technological adoption were presented to the subjects at the beginning of the experiment. We administered the quantitative effort test in the second stage. Unlike in Study 1, the effort test was now more extensive and consisted of a full section from the Psychometric Entrance Test that included 20 questions that needed to be answered within 20 min. The first two stages were meant to deepen the subjects' sense of profit from their participation (theory of mental accounting). The investment decision-making game played in the third stage now consisted of one round, which allowed us to use a direct NIS cash method and not an experiment currency. In addition, in this experiment, we did not administer a trust-type questionnaire.

For completing the questionnaires in the first stage, subjects earned NIS 10. Providing at least two correct answers on the quantitative test yielded an additional NIS 30 payment and advancement to the last stage. Those who did not respond satisfactorily were paid an additional NIS 20 and finished the experiment. In the investment decision game, subjects could earn up to an additional NIS 40.

3.2.3. Results

Analysis of the results is presented in the same manner as for Study 1. We start with RTA and follow with WTP.

3.2.3.1. Differences in RTA

Similar to in Study 1, for the total sample, we find no significant differences between the advice alternatives. However, compared with the results obtained in Study 1, here, we see significantly wider adoption of the various advice alternatives (mean RTA = 0.66 and 0.75 in Study 1 and Study 2, respectively) These results are consist with Hypotheses 1, 2, and 3. These results may be because of greater attention in laboratory than in online experiments or because of different subject characteristics. Subjects, across treatments, understood that they were being offered a service that improved their chances of receiving a better outcome and most of them wanted to use it. This leads to an RTA of over 75% (76% each for the algorithm and human advisor and 69% for the algo-human). The analysis for the subgroups (gender, age) proceeds with the same methodology performed in Study 1. The regression estimations are as in Equations (1) and (2) for the RTA analysis.¹⁹ See Table 5, Columns 1 and 2 for the between-labels results and Table 6, columns 1–3 for within-label alternatives results.

[Table 5, Table 6]

When we examine the relationship between the RTA and gender, we find, in contrast to Study 1, no significant differences that are not consistent with Hypothesis 5 (this can be

¹⁹ The results are not sensitive to different regression models such as OLS or logit. Here, each advice alternative label is estimated separately to avoid collinearity.

attributed to the experimental methodology as well as the different subject characteristics). We find that quantitative knowledge significantly increases RTA and that it stems mainly from algo-human treatment. For the technology adoption constructs, we find that those who perceived technology as effective (performance expectancy) were willing to adopt the algorithm and algo-human alternatives significantly and marginally more, respectively.

3.2.3.2. Differences in WTP

Although there is no difference in the RTA, our analysis now turns to the WTP as a measurement of trust differences for each of the three advice alternatives. Again, and as for the RTA, the analysis was conducted with the same methodology as in Study 1.²⁰ We find significant WTP differences: Subjects who were offered algorithmic advice were willing to pay more for the advice (for the same service) than subjects who were offered the human and algo-human options, NIS 3.2 (or about 30% more) and NIS 5.2 (or about 60% more) on average, respectively, for the same service. However, this inferiority is statistically significant in relation to the algo-human but not in relation to the human advisor. This is an indication of different trust levels for the different labels and provides additional support for the nonsignificant tendency documented in Study 1 for the algorithmic advice. For the technology adoption constructs, (similar to the RTA and the results in Study 1), when they perceived technology as useful (performance expectancy), subjects were willing to pay significantly more for algorithmic advice. In summary, the laboratory experiment with a student sample provides additional evidence that a younger, educated population has less aversion to and even a preference for the suggested service labeled “algorithm” compared with the suggested service labeled “human” or “algo-human” advice. These results on the RTA and WTP are partially support Hypothesis 4.

²⁰ We use the same specification as in Equations (1) and (2) only now the dependent variable is WTP as in Study 1. Results are presented in Table 5, Columns 3–5 and Table 6, Columns 4–6.

4. Studies 3 and 4: At the outbreak of COVID-19—Online experiments with a representative and a student sample

4.1. Study 3: Motivation, method, and design

Given the unexpected economic consequences and the increasing use of technology that has resulted from the global COVID-19 pandemic, the next plausible investigation is to study the effect during this unique time frame. To enable a comparison of the results to the period before COVID-19, the methodology and design were the same as in Study 1. Only now, the study was conducted as the COVID-19 pandemic was causing a financial crisis around the world. The experiment was conducted in mid-April 2020 in Israel at a time when stay-at-home orders were in effect. At this time, the S&P 500 and the Tel Aviv35 index were 17.5% and 22%, respectively, lower compared to February 19, 2020, the beginning of the market plunge.

We follow the same screening mechanism imposed for Study 1. There were 722 initial respondents recruited through Panels4All. Fifty-two participants were excluded as they did not earn a minimum grade on the quantitative effort test. An additional 144 subjects who completed the experiment in less than the minimum time of 10 min were dropped. Our final sample thus consisted of 526 subjects (mean age = 39.4 years; 51.1% male), with 178 randomly assigned to the algorithm, 178 to the algo-human, and 170 to the human advisor. See Table 7 for the sample's descriptive statistics.

[Table 7]

4.2. Study 3: Results

The results are presented as follows. We start by comparing the RTA and then the WTP between the two online, representative sample experiments (Studies 1 and 3). We then follow with the description of the results obtained for Study 3 and Study 4 alone.

4.2.1. Differences in RTA before and at the outbreak of COVID-19

First, we find that participants at the outbreak of COVID-19 were overall significantly more ready to adopt advice, by 10.6%, compared to participants before the pandemic (before COVID-19: RTA = 0.66, at the outbreak of COVID-19: RTA = 0.73), a result that supports Hypothesis 6. We next conduct a matching exercise, a propensity score matching (PSM) analysis, and a multivariate regression that enable us to control for the sample subgroups, such as gender and age and different participant characteristics.

For the PSM analysis, we use time frame (a dummy variable: 1 for at the outbreak of COVID-19 and 0 for before COVID-19), age group (the three age groups as specified in Study 1), and gender as the matching variables. For the matching specification, we use logit for the estimation and the nearest neighbors method for the PSM.

As mentioned, we find that the differences in RTA of participants at the outbreak of COVID-19 are significant, with participants wanting to use the suggested advice more before the pandemic.

For the regression analysis, we use the following specification:

$$(3) \text{ RTA}_i = \alpha + \beta \text{during}_i + \delta \text{FL}_i + \gamma X_i + \varepsilon_i$$

where the outcome variable RTA for individual i is a function of time frame (1 = during, 0 otherwise), the financial literacy index indicator (based on Lusardi and Mitchell 2011), and the control variables (quantitative knowledge, gender, religiousness, work). We find an increase in RTA at the outbreak of the pandemic that is marginally significantly higher ($p = 0.07$) and that quantitative knowledge and gender still significantly increase advice adoption ($p = 0.00$ for both), as documented in Study 1.

4.2.2. Differences in WTP before and at the outbreak of COVID-19

For the WTP analysis, we follow the same approach as for the RTA (PSM and regression analysis). The differences in the WTP between the different periods are not

significant (before COVID-19: WTP = 10.6, at the outbreak of COVID-19: WTP = 11.6).²¹ In sum, while at the outbreak of the pandemic our subjects were willing to adopt more advice, they were not willing on average to pay more for that advice.

4.2.3. Study 3: Differences in RTA at the outbreak of COVID-19

The analysis of Study 3, which was conducted at the outbreak of COVID-19 to explore the factors that affect people's choices and trust during this unique time period, follows the analysis conducted of Study 1. We start by comparing the RTA and WTP of the different manipulations and further use several relevant control characteristics to explore RTA and WTP: gender, age, trust antecedents, quantitative knowledge, financial literacy, and attitude toward technology adoption as measured by the UTAUT2. Similar to in Study 1, we observe no significant differences between the three advising alternatives in the RTA for the total sample. However, unlike Study 1's results, the lack of sensitivity to the advisor's purported type remains, even after dividing the sample into the three different age groups. That is, we no longer observe the aversion to algorithmic advice documented in Study 1 for the middle age group (31- to 44-year-olds). Participants in this age group were ready to adopt the advice labeled "algorithm" with no significant difference compared to the advice labeled human (mean RTA = 0.64, 0.75, and 0.80 for the human, algorithm, and algo-human alternatives, respectively) supporting Hypothesis 7. Figure 5 shows the means of the RTA for the different suggested advice alternatives grouped by age and gender. Table 8 show the regression results.

[Figure 5, Table 8]

When we explore gender differences, we find that men were still significantly more ready than women to adopt advice in general (RTA = 0.80 for men and 0.65 for women). This result is consistent with Study 1. The algorithm and algo-human alternatives drive the

²¹ One can relate the limited differences to the fact that there is a payment limit, since the value the advice creates equals 12 Zu for a risk-neutral individual. It seems that at the outbreak of COVID-19, the mean WTP is almost at that level.

significant differences. We find that men were still significantly more willing to adopt these alternatives than women (RTA = 0.80 and 0.81, respectively, for men and only 0.62 and 0.65, respectively, for women). Figure 5 depicts the main results. Even after controlling for other related variables using multivariate regression as specified in Study 1, we find that there are no significant differences in adopting the three advice alternatives, and men were ready to adopt advice more than women (Table 9), a result driven mainly by the algo-human alternative. We also document that quantitative knowledge, as in Studies 1 and 2, is correlated significantly with adoption (Table 9).

[Table 9]

4.2.4. Study 3: Differences in WTP at the outbreak of COVID-19

As in Study 1, we observe no significant differences between the three advising alternatives in the WTP for the total sample (Figure 5). However, unlike in Study 1, when we divide the sample into the three age groups, the lack of differences remains, meaning, as observed in the RTA, that we do not find an aversion to algorithmic advice, as seen in Study 1. Participants in the middle age group (31- to 44-year-olds) showed no significant difference in their WTP for algorithmic advice versus human advice (mean WTP = 10.2, 11.3, and 10.8 for the human, algorithm, and algo-human alternatives, respectively). Also, we do not observe the significant tendency (although the direction remains) to pay the least for the algo-human advice in the 45+ age group, as observed in Study 1 (mean WTP = 13.1, 15.1, and 11.7 for the human, algorithm, and algo-human alternatives, respectively).

When we explore gender differences, we find that men were significantly more willing to pay for advice in general (WTP = 12.8 for men and 10.2 for women). This result is consistent with Study 1. The significant differences are driven mainly by the algo-human alternative: We find that men were significantly more willing to pay for this alternative than women (WTP = 12.7 for men and only 8.5 for women). When we control for other related variables using the

multivariate regression as specified in Study 1, we find that the results hold (see Tables 8 and 9).

Overall, at the outbreak of COVID-19, participants were more ready to adopt advice and less sensitive to the advice mechanism than in pre-pandemic times. An additional explanation for the lack of algorithm aversion observed in the 31- to 44-year-olds might be that all services provided during the time of the experiment were carried out online, which may attest to preference changes and more openness to such services. The differences we find in gender are consistent.

4.3. Study 4: Online experiment with a student sample

As in the period before the COVID-19 pandemic, at the outbreak of COVID 19, we aimed to study a targeted population of students. Since a lockdown was in place, we were unable to run it in the laboratory. Hence, unlike Study 2, this experiment was conducted online. The design was adjusted to address the different environment: (1) The anonymized nature of the investigation enabled us to run the experiment without real monetary consequences or incentives. (2) As in Studies 1 and 3, the quantitative effort test was administered in the first stage and the decision-making game followed in the second stage. (3) However, here, we shortened the questionnaires after the game, adding only demographic questions with no financial literacy, technology adoption, or trust antecedents questions. We had 127 student participants, 41 randomly assigned to the algorithm treatment, 45 to the algo-human, and 41 to the human.²²

The results of Study 4 are consistent with those of Study 3. We find no significant differences between the three advising alternatives in the RTA or WTP for the total sample. Although gender differences are again significant, with men more ready to adopt advice, the difference is driven mainly by the algorithm and algo-human alternatives. Similar to in Study

²² We did not exclude any participants from the sample as they all achieved the minimum grade on the effort test.

3, we do not observe WTP differences for the advice alternatives, as the student participants were willing to pay, on average, around 12 Zu for the service (12.13 and 12.42 for men and women, respectively).

5. Conclusions

The concept of a robo-advisor financial tool, that is, an algorithmic consultant and portfolio manager, has been developing in recent years. Despite the relative rise in popularity of the tool, its use is still somewhat limited, and little is known about the reasons for adoption, WTP, and perceived trust in such tools. Moreover, some providers are advocating for the use of a hybrid model (service that is based on a combination of algorithmic and human advice). Our research employs online and laboratory investment experiments to estimate the RTA and WTP (as a proxy for trust) for a robo-advisor versus a human advisor versus a hybrid model, both before and at the outbreak of the COVID-19 pandemic when lockdown orders were imposed.

We make several contributions to the literature. We experimentally explore perceived trust in the form of WTP for the relatively new and growing service of robo-advisors and the hybrid model before and at the emergence of the COVID-19 crisis. We document that during non-pandemic times, individuals of different ages and genders differ in their preferences and trust in the different types of financial advice. One can conjecture that the age differences obtained before COVID-19 either are due to age-related conceptual differences in the desire to explore innovation or reflect an intergenerational change in the exposure to technology.

At the outbreak of the COVID-19 crisis, participants were willing to accept more advice and were less sensitive to the suggested advice type, although gender differences remain significant. The absence of algorithm aversion for the middle age group at the outbreak of the COVID-19 pandemic might be the result of more openness to technology due to the digitization of most services provided during the pandemic lockdown. We leave for future research to

investigate if this reduction in algorithm aversion remains once we overcome the COVID-19 challenges and more travel and human interaction are enabled again. The gender differences that we document in our study regardless of the investigated period are puzzling. While this result is consistent with several studies in the literature on technology adoption and gender, it is less consistent with our expectations based on the financial literacy, overconfidence, risk aversion, and gender literature.

Further, we use different experimental methods (online vs. laboratory) and discuss the advantages and disadvantages of each method. We use different types of subjects (representative sample and students) to enrich the investigation.

Our approach in this project is multidisciplinary and we use accepted measures from different disciplines. We contribute to the information and technology literatures investigating the different factors that affect adoption of algorithm-driven services in various areas of life. Although we focus on the use of financial robo-advisors, our method of investigation and our findings are related to many other potential areas that investigate adoption of algorithms as decision makers. Specifically, we find that using an estimation method that adds the WTP to the common measure of RTA questionnaires enriches the discussion and provides additional important insights.

We contribute to the recent growing literature that investigates economic issues related to the COVID-19 pandemic, which is clearly a pandemic that is far more than just a health crisis. We leave for future research further investigation of the adoption of advanced financial technology services and the COVID-19 pandemic.

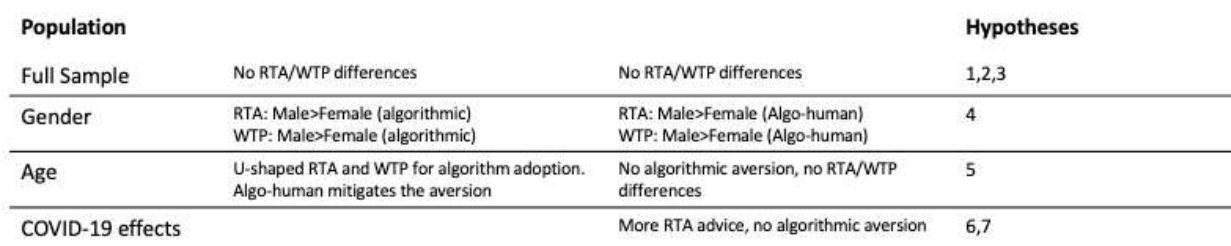
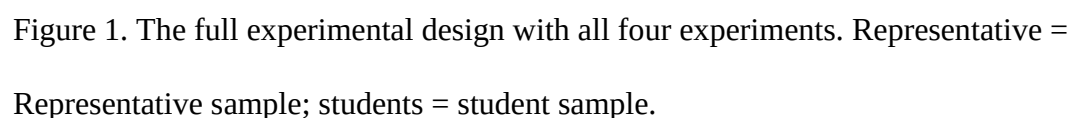
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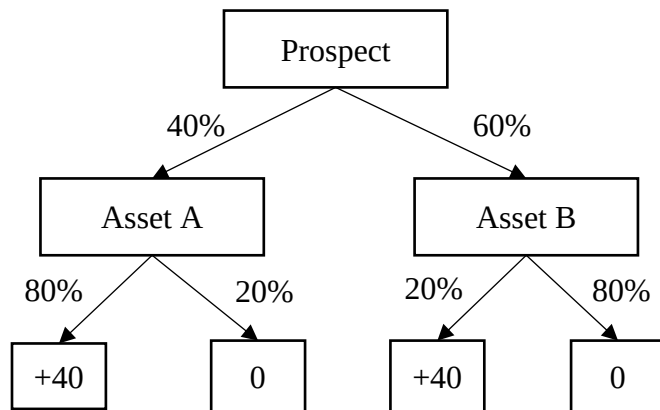
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A



B

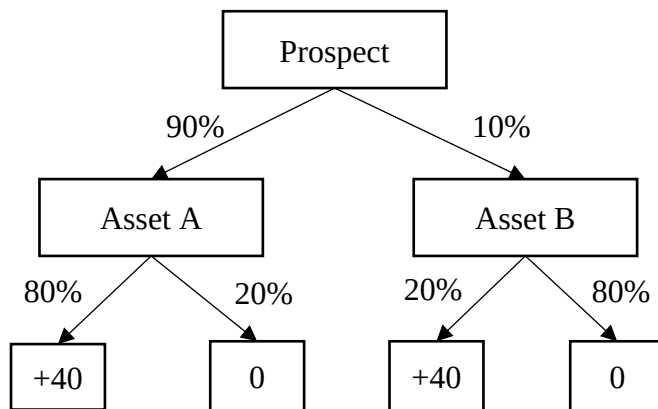


Figure 3. (A) The initial probability structure presented to subjects and (B) the probability structure after subjects were offered an agent investment service, in Study 1.

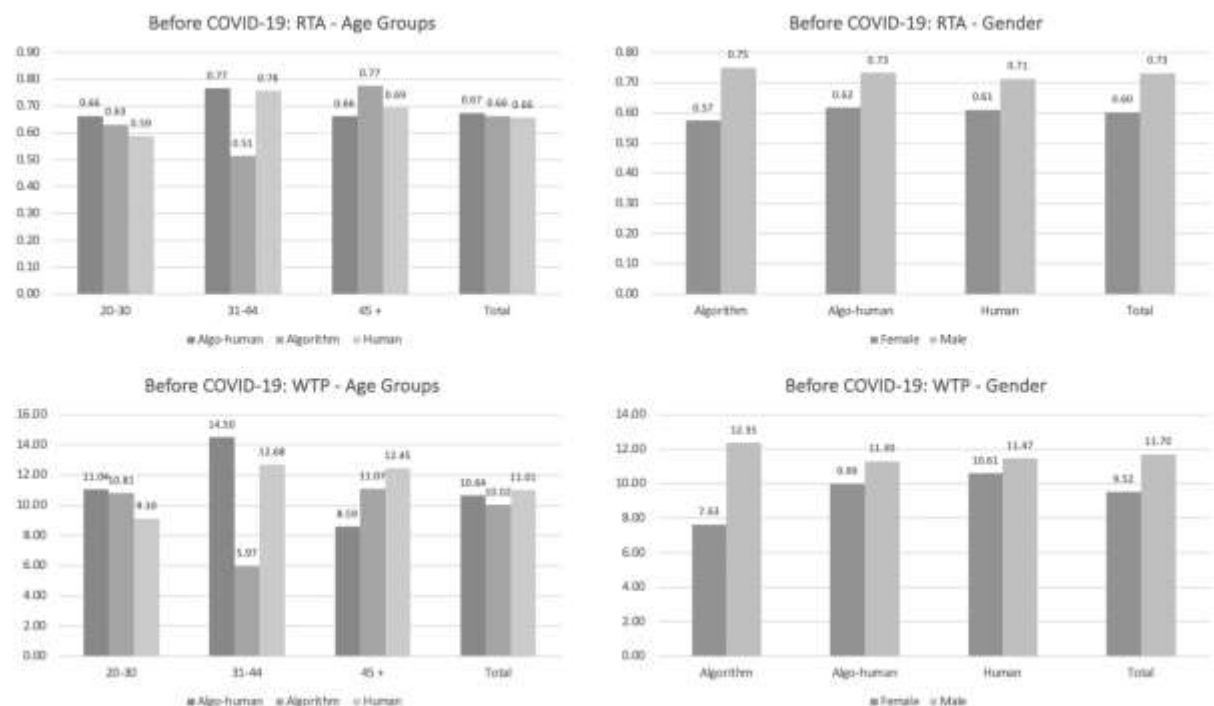


Figure 4. Study 1: Before COVID-19 pandemic. Mean readiness to adopt (RTA) and mean willingness to pay (WTP) for investment advice in the total sample and for each treatment: algorithm, algo-human (algorithm with human assistance), and human. The data are grouped by age group (in years; left) and gender (right).

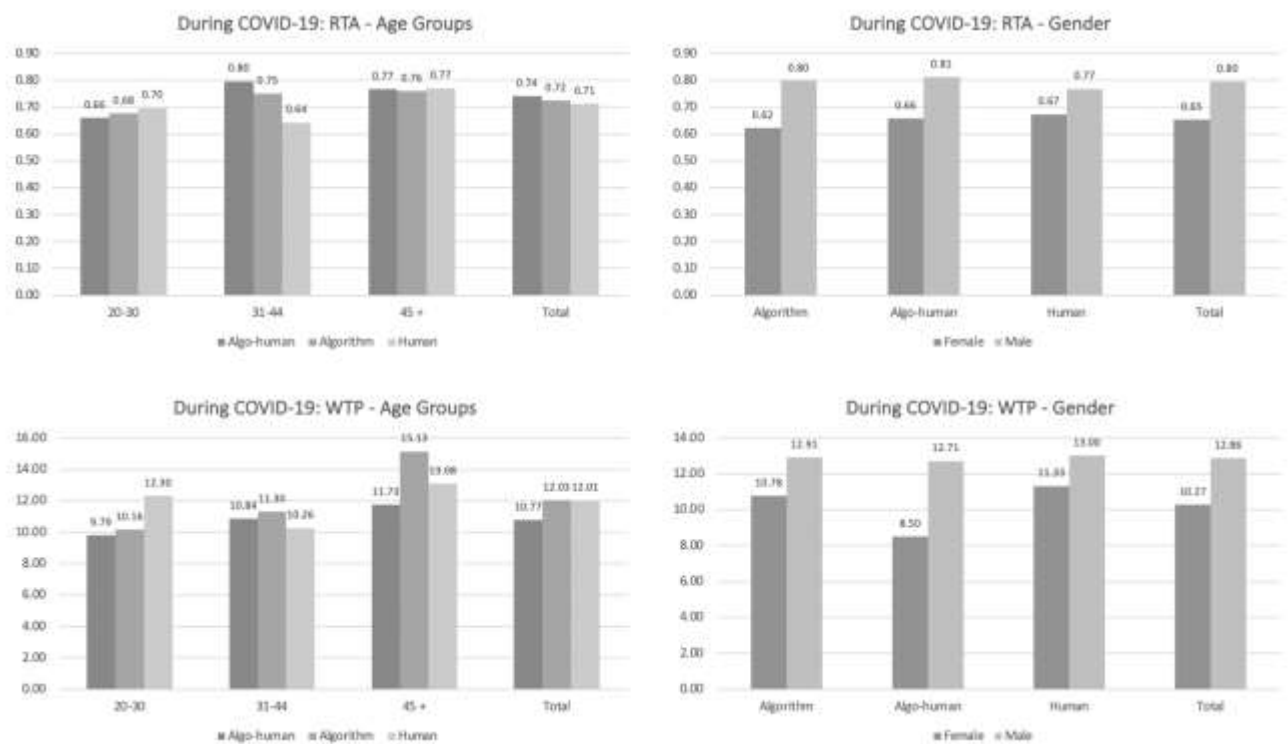


Figure 5. Study 3: At the outbreak of the COVID-19 pandemic. Mean readiness to adopt (RTA) and mean willingness to pay (WTP) for investment advice in the total sample and for each treatment: algorithm, algo-human (algorithm with human assistance), and human. The data are grouped by age group (left) and gender (right).

Table 1. Study 1 descriptive statistics

Variable	Advice treatment			Total sample
	Human	Algorithm	Algo-human	
WTP, Round 3	11.01 (12.42)	10.02 (10.70)	10.64 (12.20)	10.58 (11.82)
RTA, Round 3	0.658 (0.476)	0.663 (0.474)	0.675 (0.470)	0.665 (0.472)
Age (years)	38.32 (13.61)	37.74 (11.88)	36.70 (13.27)	37.60 (12.98)
Gender (% male)	0.461 (0.500)	0.505 (0.501)	0.498 (0.501)	0.487 (0.500)
Work	0.813 (0.391)	0.895 (0.308)	0.828 (0.379)	0.843 (0.364)
Quantitative knowledge	2.498 (1.175)	2.342 (1.257)	2.315 (1.090)	2.389 (1.175)
<i>N</i>	219	190	203	612

Notes: Means and standard errors (in parentheses) unless otherwise indicated. For parsimony, work is a dummy variable (work = full- or part-time employment). Quantitative knowledge = The number of correct answers on the psychometric test; WTP = willingness to pay; RTA = readiness to adopt.

Table 2. Study 1: Results of the probit analysis for readiness to adopt (RTA) and the tobit analysis for willingness to pay (WTP) for the total sample and by age group

Variable	RTA: Probit				WTP: Tobit			
	Total	20–30 years	31–44 years	45+ years	Total	20–30 years	31–44 years	45+ years
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Algorithm	0.0354 (0.132)	0.127 (0.197)	-0.729** (0.336)	0.277 (0.223)	-0.872 (1.173)	1.756 (1.824)	-7.025** (2.734)	-1.127 (1.769)
Algo-human	0.0583 (0.129)	0.214 (0.187)	-0.0118 (0.367)	-0.0850 (0.214)	-0.268 (1.146)	1.967 (1.730)	1.370 (2.853)	-3.662** (1.765)
Quantitative knowledge	0.137*** (0.0474)	0.165** (0.0675)	0.295** (0.145)	0.105 (0.0816)	0.845** (0.413)	1.126* (0.617)	1.289 (1.082)	0.627 (0.647)
Financial literacy	0.0929 (0.0618)	0.0462 (0.0914)	0.378** (0.168)	-0.0148 (0.116)	-0.0161 (0.555)	-0.468 (0.847)	2.010 (1.341)	-0.447 (0.936)
Gender: Male	0.292*** (0.111)	0.291* (0.159)	0.0836 (0.291)	0.385** (0.193)	2.005** (0.985)	3.074** (1.464)	-1.728 (2.374)	2.024 (1.560)
Work	-0.259* (0.152)	-0.321 (0.216)	0.310 (0.492)	-0.414 (0.260)	-0.867 (1.313)	-0.824 (1.914)	1.507 (4.123)	-2.072 (1.990)
Religiousness	-0.138 (0.112)	-0.170 (0.160)	0.0544 (0.294)	-0.0138 (0.200)	0.0645 (1.004)	-0.856 (1.489)	2.325 (2.442)	1.359 (1.619)
Constant	0.0152 (0.210)	-0.0865 (0.307)	-0.924 (0.601)	0.466 (0.363)	8.684*** (1.878)	6.664** (2.861)	4.872 (4.919)	12.26*** (2.889)
<i>N</i>	612	282	102	227	612	282	102	227

Notes: Probit and tobit regression coefficients; standard errors are in parentheses. Dependent variables are individual *i*'s RTA and WTP. The main explanatory variables are the algorithm and algo-human treatment indicators compared to human. Religiousness and work variables are dummies (religiousness = perceives self as religious to some extent; work = full- or part-time employment); financial literacy: 1–3 indicator of number of correct answers; quantitative knowledge = the number of correct answers on the psychometric test.
 * $p < 0.10$. ** $p < 0.05$. *** $p < 0.01$.

Table 3. Study 1: Results of the probit analysis for readiness to adopt (RTA) and the tobit analysis for willingness to pay (WTP) for the total sample and by advice treatment with technology adoption and trust antecedent control constructs

Variable	RTA: Probit				WTP: Tobit (with technology adoption)				WTP: Tobit (with trust antecedents)		
	Total	Human	Algorithm	Algo-human	Total	Human	Algorithm	Algo-human	Human	Algorithm	Algo-human
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Quantitative knowledge	0.131*** (0.0479)	0.0681 (0.0801)	0.150* (0.0861)	0.202** (0.0946)	0.911** (0.414)	0.493 (0.738)	0.251 (0.609)	1.901** (0.784)	0.113 (0.707)	0.214 (0.580)	1.436* (0.755)
Financial literacy	0.0631 (0.0635)	0.0285 (0.105)	0.194 (0.123)	-0.0109 (0.117)	-0.0178 (0.564)	1.317 (0.989)	-0.105 (0.924)	-1.311 (0.969)	1.010 (0.937)	-0.223 (0.866)	-1.776* (0.924)
Gender: Male	0.332*** (0.114)	0.275 (0.197)	0.594*** (0.217)	0.287 (0.202)	2.116** (1.009)	0.174 (1.820)	5.520*** (1.615)	0.769 (1.724)	-0.410 (1.714)	3.203** (1.531)	1.456 (1.657)
Work	-0.275* (0.154)	-0.118 (0.235)	-0.987** (0.414)	-0.184 (0.270)	-1.269 (1.312)	-0.988 (2.130)	-1.995 (2.496)	-0.922 (2.209)	0.00130 (2.062)	-0.852 (2.347)	-0.643 (2.128)
Religiousness	-0.119 (0.113)	0.263 (0.190)	-0.401* (0.217)	-0.386* (0.204)	0.104 (1.005)	0.958 (1.735)	0.294 (1.610)	-1.135 (1.766)	0.663 (1.656)	1.006 (1.587)	-0.922 (1.706)
Technology adoption											
Performance expectancy	0.0653 (0.0875)	0.0356 (0.148)	0.367* (0.191)	-0.0114 (0.157)	1.047 (0.772)	1.049 (1.382)	2.928** (1.307)	-0.0650 (1.323)			
Effort expectancy	-0.118 (0.0802)	-0.136 (0.140)	0.179 (0.169)	-0.229 (0.143)	-1.497** (0.703)	-0.632 (1.295)	0.482 (1.225)	-1.943* (1.163)			
Facilitating conditions	0.0882 (0.0829)	0.219 (0.136)	-0.379** (0.167)	0.181 (0.158)	0.248 (0.729)	1.581 (1.257)	-2.185* (1.232)	-0.533 (1.348)			
Hedonic motivation	0.0130 (0.0797)	-0.000345 (0.139)	-0.175 (0.161)	0.0824 (0.136)	0.461 (0.708)	-0.405 (1.283)	-1.364 (1.151)	2.528** (1.204)			
Experience and habit	-0.104* (0.0567)	-0.0916 (0.0910)	-0.111 (0.114)	-0.0805 (0.103)	0.153 (0.494)	0.199 (0.843)	-0.288 (0.841)	0.905 (0.854)			
Social influence											

Variable	RTA: Probit				WTP: Tobit (with technology adoption)				WTP: Tobit (with trust antecedents)		
	Total	Human	Algorithm	Algo-human	Total	Human	Algorithm	Algo-human	Human	Algorithm	Algo-human
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Behavioral intention	0.0203 (0.0504)	0.0189 (0.0819)	0.0190 (0.0947)	-0.0106 (0.0962)	0.347 (0.439)	0.443 (0.749)	0.346 (0.693)	0.0764 (0.817)			
	0.0975 (0.0953)	-0.0165 (0.172)	0.0528 (0.188)	0.258 (0.179)	-0.319 (0.836)	-2.166 (1.576)	0.980 (1.331)	0.0607 (1.469)			
Trust antecedent											
Anxiety									0.650 (0.683)	1.346** (0.592)	-0.0142 (0.690)
Calculational: Integrity									0.808 (1.212)	-0.131 (1.145)	-0.485 (1.387)
Calculational: Competence									1.251 (1.114)	-0.697 (1.010)	1.247 (1.087)
Calculational: Emotional									1.603** (0.716)	1.346** (0.650)	2.012** (0.823)
Institutional: Structural									-2.083** (0.917)	0.429 (0.794)	0.232 (0.998)
Constant	-0.360 (0.373)	-0.193 (0.630)	0.406 (0.783)	-0.920 (0.649)	6.227* (3.296)	7.661 (5.855)	2.241 (5.616)	5.243 (5.491)	-2.468 (4.107)	-0.662 (4.104)	-1.985 (4.342)
N	612	219	190	203	612	219	190	203	219	190	203

Notes: Probit and tobit regression coefficients; standard errors are in parentheses for each treatment. Dependent variables are individual *i*'s RTA and WTP for the different advice alternatives. Religiousness = perceives self as religious to some extent; work = full- or part-time employment; financial literacy: 1–3 indicator of number of correct answers (see questions in Appendix D); quantitative knowledge = the number of correct answers on the psychometric test; technology adoption and trust antecedents rated on scales of 1–7 (see questions in Appendixes B and C). Technology adoption constructs: performance expectancy = the belief in how beneficial the technology will be; effort expectancy = how easy it is to use; facilitating conditions = what resources and support are available; hedonic motivation = degree of pleasure; experience and habit = acting automatically through learning; social influence = what one's important others think; behavioral intention = the intent to try new technologies. Trust antecedents: Anxiety = distrust; calculational = trust based on the rational explanations from all available information generated; institutional = environmental and not direct trust.

* $p < 0.10$. ** $p < 0.05$. *** $p < 0.01$.

Table 4. Study 2 descriptive statistics

Variable	Advice treatment			Total sample
	Algorithm	Algo-human	Human	
Age (years)	23.31	23.06	22.10	22.8
Gender (% male)	0.35	0.42	0.24	0.38
WTP	13.76	8.54	10.54	10.86
RTA (%)	0.76	0.69	0.76	0.75
Work	0.41	0.58	0.54	0.51
	(0.07)	(0.07)	(0.07)	(0.04)
Education status (% undergraduate)	0.96	0.96	0.96	0.96
Quantitative knowledge	12.59	12.98	13.50	12.98
	(0.45)	(0.49)	(0.38)	(3.17)
<i>N</i>	51	52	50	153

Notes: Means and standard errors (in parentheses). For parsimony, the work variable is a dummy (work = full- or part-time employment). Quantitative knowledge = The number of correct answers on the psychometric test; RTA = readiness to adopt; WTP = willingness to pay.

Table 5. Study 2: Results of the probit analysis for readiness to adopt (RTA) and the tobit analysis for willingness to pay (WTP)

Variable	RTA: Probit			WTP: Tobit	
	(1)	(2)	(3)	(4)	(5)
Algorithm	0.199 (0.299)		5.159** (2.308)		
Algo-human		-0.0281 (0.287)		-1.651 (2.039)	-4.722** (2.142)
Financial literacy	0.147 (0.182)	0.00704 (0.178)	0.409 (1.398)	-0.812 (1.267)	-0.265 (1.287)
Quantitative knowledge	0.0124 (0.0486)	0.0844* (0.0477)	0.741* (0.392)	0.568* (0.332)	0.527 (0.333)
Gender: Male	0.122 (0.319)	0.146 (0.320)	0.705 (2.386)	0.297 (2.195)	-2.413 (2.230)
Age	-0.00336 (0.0366)	-0.0142 (0.0630)	-0.236 (0.230)	-0.0856 (0.439)	-0.0530 (0.213)
Religiousness	-0.140 (0.301)	-0.586* (0.304)	-0.800 (2.289)	1.382 (2.081)	-0.433 (2.126)
Work	0.729** (0.333)	-0.211 (0.323)	5.334** (2.427)	-0.955 (2.292)	-1.410 (2.286)
Constant	-0.0154 (1.012)	0.244 (1.409)	11.14*** (0.780)	9.939*** (0.692)	10.64*** (0.738)
<i>N</i>	101	102	102	103	104

Notes: Probit and tobit regression coefficients; standard errors are in parentheses. Dependent variables are individual *i*'s RTA and WTP. The main explanatory variables are the algorithm and the algo-human treatment indicators where the base alternatives are human (columns 1–4) and algorithm (column 5) in that order. For parsimony, religiousness and work variables are dummies (religiousness = perceives self as religious to some extent; work = full- or part-time employment). Financial literacy based on Lusardi and Mitchell (2011): 1–3 indicator of number of correct answers. Quantitative knowledge = The number of correct answers on the psychometric test.

Table 6. Study 2: Results of the probit analysis for readiness to adopt (RTA) and the tobit analysis for willingness to pay (WTP) by advice treatment, with technology adoption constructs

Variable	RTA: Probit			WTP: Tobit		
	Algorithm	Algo-human	Human	Algorithm	Algo-human	Human
	(1)	(2)	(3)	(4)	(5)	(6)
Financial literacy	0.156 (0.373)	0.129 (0.342)	0.210 (0.380)	0.0437 (1.990)	0.178 (1.536)	0.572 (2.192)
Quantitative knowledge	0.157 (0.0989)	0.249** (0.111)	-0.0882 (0.102)	0.904* (0.526)	0.539 (0.365)	0.839 (0.524)
Gender: Male	-0.0769 (0.755)	-0.939 (0.728)	1.641* (0.893)	-0.417 (3.606)	-3.005 (2.730)	5.380 (3.710)
Age	0.604* (0.319)	0.125 (0.129)	-0.401** (0.199)	-0.255 (0.248)	0.245 (0.493)	-0.507 (0.779)
Religiousness	1.580 (1.093)	-0.560 (0.591)	-1.052 (0.653)	-4.220 (3.163)	1.454 (2.444)	0.453 (3.422)
Work	-0.216 (0.773)	-2.165** (0.876)	1.231** (0.620)	6.802* (3.638)	-5.153* (2.934)	2.715 (3.487)
Technology adoption						
Performance expectancy	1.300*** (0.491)	1.113* (0.654)	0.200 (0.434)	5.222*** (1.835)	3.093 (2.332)	1.749 (2.432)
Effort expectancy	-0.268 (0.612)	-0.672* (0.349)	-0.206 (0.426)	-1.004 (2.646)	-0.399 (1.459)	-3.621 (2.532)
Social influence	0.257 (0.338)	-0.00611 (0.234)	-0.178 (0.240)	-2.030 (1.273)	-0.725 (1.140)	-2.828** (1.197)
Facilitating conditions	-0.304 (0.484)	0.250 (0.427)	-0.0494 (0.347)	1.357 (2.419)	-2.396 (1.758)	1.813 (2.212)
Hedonic motivation	-0.737* (0.428)	-0.302 (0.365)	0.256 (0.350)	-3.251 (1.984)	-0.703 (1.453)	0.832 (1.885)

Variable	RTA: Probit			WTP: Tobit		
	Algorithm	Algo-human	Human	Algorithm	Algo-human	Human
	(1)	(2)	(3)	(4)	(5)	(6)
Experience and habit	0.272 (0.331)	0.0433 (0.273)	-0.124 (0.365)	1.531 (1.728)	1.231 (1.154)	3.292* (1.634)
Constant	-17.71* (9.602)	-6.557* (3.467)	9.982 (6.082)	10.52*** (1.042)	7.987*** (0.783)	9.569** * (0.957)
N	51	52	50	51	52	50

Notes: Probit and tobit regression coefficients; standard errors are in parentheses for each treatment. Dependent variables are individual i 's RTA and WTP. The main explanatory variables are the algorithm, algo-human, and human treatment indicators. For parsimony, religiousness and work variables are dummies (religiousness = perceives self as religious to some extent; work = full- or part-time employment). Financial literacy based on Lusardi and Mitchell: 1–3 indicator of number of correct answers (see questions in Appendix D). Quantitative knowledge = The number of correct answers on the psychometric test. Technology adoption constructs: performance expectancy = the belief in how beneficial the technology will be; effort expectancy = how easy it is to use; social influence = what one's important others think; facilitating conditions = what resources and support are available; hedonic motivation = degree of pleasure; experience and habit = acting automatically through learning (see questions in Appendix C). Unreported control: education status (96% were undergraduates).

Table 7. Study 3 descriptive statistics

Variable	Advice treatment			Total sample
	Human	Algorithm	Algo-human	
WTP, Round 3	12.01 (11.54)	12.03 (12.27)	10.77 (10.56)	11.60 (11.47)
RTA, Round 3	0.712 (0.454)	0.725 (0.448)	0.742 (0.439)	0.726 (0.446)
Age (years)	39.57 (15.92)	38.51 (14.62)	40.07 (14.80)	39.38 (15.10)
Gender (% male)	0.406 (0.493)	0.584 (0.494)	0.539 (0.500)	0.511 (0.500)
Work	0.753 (0.433)	0.747 (0.436)	0.775 (0.419)	0.759 (0.428)
Quantitative knowledge	2.447 (1.120)	2.612 (1.236)	2.826 (1.248)	2.631 (1.211)
<i>N</i>	170	178	178	526

Notes: Means and standard errors (in parentheses) unless otherwise indicated. For parsimony, work variable is a dummy (work = full- or part-time employment). Quantitative knowledge = The number of correct answers on the psychometric test; WTP = willingness to pay; RTA = readiness to adopt.

Table 8. Study 3: Results of the probit analysis for readiness to adopt (RTA) and the tobit analysis for willingness to pay (WTP) for the total sample and by age group

Variable	RTA: Probit				WTP: Tobit			
	Total	20–30 years	31–44 years	45+ years	Total	20–30 years	31–44 years	45+ years
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Algorithm	-0.0612 (0.148)	-0.129 (0.234)	0.191 (0.281)	-0.138 (0.277)	-0.373 (1.233)	-2.011 (1.984)	0.539 (2.184)	1.978 (2.203)
Algo-human	-0.0286 (0.149)	-0.102 (0.249)	0.194 (0.288)	-0.210 (0.267)	-1.458 (1.235)	-2.297 (2.114)	-0.195 (2.208)	-1.705 (2.128)
Quantitative knowledge	0.153*** (0.0526)	0.0666 (0.0876)	0.234** (0.101)	0.274*** (0.100)	0.00343 (0.429)	-0.589 (0.734)	0.557 (0.749)	0.465 (0.752)
Financial literacy	0.0365 (0.0677)	-0.122 (0.110)	0.150 (0.131)	0.108 (0.125)	-0.689 (0.570)	-1.756* (0.933)	-1.020 (1.006)	0.867 (1.046)
Gender: Male	0.407*** (0.125)	0.670*** (0.203)	0.104 (0.235)	0.294 (0.227)	3.034*** (1.040)	4.993*** (1.708)	2.913 (1.781)	0.0822 (1.845)
Work	-0.00578 (0.139)	-0.0487 (0.224)	0.145 (0.318)	-0.0787 (0.242)	0.559 (1.162)	2.076 (1.893)	2.020 (2.494)	-1.728 (1.945)
Religiousness	-0.0844 (0.121)	-0.230 (0.209)	0.245 (0.238)	0.0228 (0.239)	-1.001 (1.010)	-1.243 (1.757)	-0.120 (1.792)	1.255 (1.927)
Constant	0.0176 (0.221)	0.473 (0.378)	-0.693 (0.450)	-0.149 (0.381)	12.13*** (1.877)	13.86*** (3.201)	8.262** (3.499)	10.91*** (3.145)
<i>N</i>	526	187	162	175	526	187	162	175

Notes: Probit and tobit regression coefficients; standard errors are in parentheses. Dependent variables are individual *i*'s RTA and WTP. The main explanatory variables are the algorithm and algo-human treatment indicators compared to human. Religiousness = perceives self as religious to some extent; work = full- or part-time employment; financial literacy: 1–3 indicator of number of correct answers; quantitative knowledge = the number of correct answers on the psychometric test.

* $p < 0.10$. ** $p < 0.05$. *** $p < 0.01$.

Table 9. Study 3: Results of the probit analysis for readiness to adopt (RTA) and the tobit analysis for willingness to pay (WTP) for the total sample and by advice treatment with technology adoption and trust antecedent control constructs

Variable	RTA: Probit				WTP: Tobit (with technology adoption)				WTP: Tobit (with trust antecedents)		
	Total	Human	Algorithm	Algo-human	Total	Human	Algorithm	Algo-human	Human	Algorithm	Algo-human
Quantitative knowledge	0.151*** (0.0541)	0.161 (0.104)	0.181* (0.0978)	0.197** (0.0993)	-0.119 (0.429)	0.128 (0.823)	0.0718 (0.752)	0.00819 (0.694)	-0.177 (0.797)	-0.0358 (0.732)	-0.377 (0.666)
Financial literacy	0.0223 (0.0694)	0.114 (0.112)	0.148 (0.133)	-0.209 (0.140)	-0.691 (0.570)	-0.446 (0.926)	0.321 (1.087)	-1.415 (0.964)	-0.457 (0.893)	-0.0952 (1.063)	-1.833* (0.948)
Gender: Male	0.393*** (0.126)	0.183 (0.230)	0.382 (0.240)	0.618*** (0.225)	2.691*** (1.029)	2.033 (1.858)	0.327 (1.993)	4.912*** (1.619)	1.662 (1.801)	1.025 (1.867)	3.951** (1.609)
Work	-0.0316 (0.141)	-0.152 (0.259)	0.104 (0.261)	-0.0188 (0.258)	0.468 (1.144)	-1.356 (2.061)	1.484 (2.080)	1.429 (1.872)	-1.345 (1.998)	-0.0682 (2.038)	1.539 (1.835)
Religiousness	-0.0714 (0.123)	0.0337 (0.219)	-0.184 (0.228)	-0.112 (0.221)	-0.865 (1.002)	-1.054 (1.761)	-0.939 (1.849)	-1.034 (1.611)	-1.386 (1.768)	-2.596 (1.792)	-1.020 (1.572)
Technology adoption											
Performance expectancy	-0.240** (0.100)	-0.320* (0.183)	-0.235 (0.189)	-0.233 (0.195)	-1.924** (0.820)	-1.586 (1.519)	-2.326 (1.441)	-2.443* (1.363)			
Effort expectancy	-0.0953 (0.0855)	0.0251 (0.143)	-0.258 (0.182)	-0.0657 (0.151)	-0.593 (0.698)	-0.588 (1.165)	-1.406 (1.408)	0.329 (1.099)			
Facilitating conditions	0.183* (0.0962)	0.265 (0.168)	0.0301 (0.190)	0.173 (0.172)	0.273 (0.794)	0.348 (1.375)	0.217 (1.565)	0.0349 (1.283)			
Hedonic motivation	0.222** (0.0925)	0.103 (0.165)	0.537*** (0.180)	0.0630 (0.171)	2.402*** (0.744)	2.860** (1.318)	2.292* (1.339)	1.780 (1.237)			
Experience and habit	-0.0669 (0.0612)	-0.0625 (0.109)	-0.230* (0.123)	0.0462 (0.105)	-0.889* (0.489)	-0.872 (0.878)	-1.386 (0.916)	-0.365 (0.768)			

Variable	RTA: Probit				WTP: Tobit (with technology adoption)				WTP: Tobit (with trust antecedents)		
	Total	Human	Algorithm	Algo-human	Total	Human	Algorithm	Algo-human	Human	Algorithm	Algo-human
Social influence	-0.00388 (0.0524)	0.0616 (0.0856)	0.0793 (0.0917)	-0.172 (0.115)	0.374 (0.427)	0.0563 (0.707)	0.942 (0.746)	0.377 (0.799)			
Behavioral intention	0.0543 (0.110)	-0.00442 (0.215)	0.299 (0.202)	0.0583 (0.201)	0.882 (0.907)	0.852 (1.756)	2.982* (1.590)	-0.536 (1.478)			
Trust antecedent Anxiety									-0.227 (0.709)	1.404** (0.687)	0.194 (0.648)
Calculational: Integrity									-1.849 (1.591)	0.515 (1.448)	-0.640 (1.392)
Calculational: Competence									0.363 (1.111)	-0.129 (1.199)	1.075 (1.058)
Calculational: Emotional									1.758* (0.912)	2.308*** (0.845)	1.219 (0.810)
Institutional: Structural									1.540 (1.046)	-1.538 (0.983)	-0.0647 (0.888)
Constant	-0.229 (0.414)	-0.250 (0.769)	-1.622** (0.784)	1.043 (0.765)	9.000*** (3.416)	7.060 (6.266)	3.085 (6.283)	15.49*** (5.456)	7.326* (4.278)	1.410 (4.579)	4.519 (4.051)
N	526	170	178	178	526	170	178	178	170	178	178

Notes: Probit and tobit regression coefficients; standard errors are in parentheses for each treatment. Dependent variables are individual *i*'s RTA and WTP for the different advice alternatives. Religiousness = perceives self as religious to some extent; work = full- or part-time employment; financial literacy: 1–3 indicator of number of correct answers (see questions in Appendix D); quantitative knowledge = the number of correct answers on the psychometric test; technology adoption and trust antecedents rated on scales of 1–7 (see questions in Appendices C and E). Technology adoption constructs: performance expectancy = the belief in how beneficial the technology will be; effort expectancy = how easy it is to use; facilitating conditions = what resources and support are available; hedonic motivation = degree of pleasure; experience and habit = acting automatically through learning; social influence = what one's important others think; behavioral intention = the intent to try new technologies. Trust antecedents: Anxiety = distrust; calculational = trust based on the rational explanations from all available information generated; institutional = environmental and not direct trust.

* $p < 0.10$. ** $p < 0.05$. *** $p < 0.01$.

Appendix A: The Investment Game²³

In this part of the experiment, you can gain additional “Zuzims” (Zu).

Three rounds of a decision-making game will be presented to you one by one. The amount of Zu you will earn at the end of this part of the experiment will be the amount you earned in one of the three rounds that will be selected randomly.

Round 1 of the decision game

The sum of the Zu you earn in the game will be determined by two lotteries. In the first stage there will be a lottery between two possible outcomes—Result A with a probability of 40%, and Result B with a probability of 60%. If the result in the first draw is A, in the second stage you will participate in Lottery A, where you have an 80% chance of earning Zu 40 and a 20% chance of earning nothing. If the result in the first lottery is B, in the second stage you will participate in Lottery B, where there is an 80% chance of earning nothing and a 20% chance of earning Zu 40.

In other words, the structure of the current stage is as follows: (See Figure 3A)

Before we begin the lotteries, we offer you the use of the services of a completely computerized algorithm²⁴ of the insurance agency “ABC” that will advise you.

²³ The investment game was presented to the subjects in Hebrew.

²⁴ As presented for the algorithmic advice. For the algo-human hybrid alternative: “The agency has a completely computerized algorithm that can advise you on this issue, and assume that you will also have the option of consulting with a personal advisor at the agency who will explain the algorithm.” For the human alternative: “An investment advisor at insurance agency “ABC.” The subjects were allocated randomly to the alternatives.

The agency has a “completely computerized algorithm” that can “advise” you on this issue.

The algorithm²⁵ has the ability to change the first lottery, so the chances of participating in the more profitable Lottery A will increase from 40% to 90%, and the chances of participating in Lottery B will drop from 60% to 10%.

The service does not guarantee that you will win anything in the lottery but only gives advice to improve your chances of winning the more profitable lottery.

That is, if you use the service, the structure of the current phase will be as follows: (See Figure 3B)

If you are willing to use the service, you must pay a fee for it. You can offer any amount you choose from your earnings in the previous parts of the study.

If your bid is lower than the fee requested by the company for the service, no fee will be charged, and you will not receive the service.

If your bid is higher than or equal to the commission requested by the company, the company will accept the offer and provide you with the service.

²⁵ For the algorithmic advice and for the algo-human advice. “Advisor” for the human advice.

Note: Even if your offer is higher than the required fee (the value will be set randomly from a uniform distribution in a particular area), the company will charge you only the amount it charges for the service.

That is, in any case, you should offer the true value of the service to you: If this value is lower than the company fee for service, the service is expensive for you—you will not receive the service and will not pay anything.

If the company's commission is cheaper than (or identical to) the price you quote, you will pay only the fee that the company requests.

The payment for the service will be deducted from the payment to be paid to you at the end of the experiment and then the two lotteries will be held.

Go to the next screen to decide whether to use the service and if so, how much you are prepared to pay for it.

Would you like to use the service? (Yes/No)

What is the amount you are willing to pay for the service?²⁶

²⁶ The second and third rounds were conducted in a similar manner.

Appendix B: The Trust-Antecedent Questionnaire

1. The advice offered to me is capable of assisting me.
2. I feel secure conducting business with the advice offered to me because regulations in my country will protect me.
3. I feel safe using the advice offered to me because of the service provider institute.
4. I feel comfortable relying on the advice offered to me.
5. I feel the advisor has good knowledge about the advice.
6. I consider the advisor to have integrity.
7. The advice offered to me has nothing to gain by being dishonest with me.
8. The advice offered to me has nothing to gain by not caring about me.
9. The advice offered to me has nothing but the client's interest.
10. I feel safer that there is an advisor that is making this financial decision with me.

Notes: The questionnaire was based on McKnight et al. (2002), Gefen et al. (2003), and Komiak and Benbasat (2006). It was translated and presented to the subjects in Hebrew.

Appendix C: The Technology Adoption Conceptualization and Questionnaire

Performance expectancy is defined as the degree to which an individual believes that using technology will provide benefits for him or her. *Effort expectancy* is the degree to which an individual believes that technology is easy to use. *Social influence* is the degree to which an individual believes that others important to him or her (e.g., family, friends) think he or she should use the technology. *Facilitating conditions* refer to the degree to which an individual believes that resources and support are available to make use of a particular technology. *Hedonic motivation* is defined as the degree to which an individual believes that fun or pleasure will be derived from using a technology. *Experience and habit* is defined as the degree to which an individual performs behaviors automatically through learning. *Price value* is the consumer's cognitive trade-off between the perceived benefits of the technology and the monetary cost of using it (Dodds, Monroe, and Grewal 1991). In our experimental framework, we base our design on the original UTAUT2 constructs, but we use WTP as the dependent outcome variable and not as an explanatory variable, as in the original framework. The questionnaire is as follows:

Performance Expectancy

1. I find technology useful in my daily life.
2. Using technology increases my chances of achieving things that are important to me. (dropped)
3. Using technology helps me accomplish things more quickly.
4. Using technology increases my productivity.

Effort Expectancy

5. Learning how to use technology is easy for me.
6. My interaction with technology is clear and understandable.
7. I find technology easy to use.
8. It is easy for me to become skillful at using technology.

Social Influence

- 9. People who are important to me think that I should use mobile Internet.
- 10. People who influence my behavior think that I should use technology.
- 11. People whose opinions I value prefer that I use mobile Internet.

Facilitating Conditions

- 12. I have the resources necessary to use technology.
- 13. I have the knowledge necessary to use technology.
- 14. I can get help from others when I have difficulties using new technology.

Hedonic Motivation

- 15. Using technology is fun.
- 16. Using technology is enjoyable.
- 17. Using technology is very entertaining.

Experience and Habit

- 18. The use of technology has become a habit for me.
- 19. I am addicted to using technology.
- 20. I must use technology.

Behavioral Intention

- 21. I intend to continue using technology in the future.
- 22. I will always try to use technology in my daily life.
- 23. I plan to continue to use technology frequently.

Notes: The questionnaire was based on the Unified Theory of Acceptance and Use of Technology 2 (Venkatesh et al. 2012) without the price value construct that we examine with the willingness to pay. It was translated and presented to the subjects in Hebrew.

Appendix D: The Objective Financial Literacy Questionnaire

Financial literacy may play an important role in the desire to hire a financial advisor, yet it has a relatively ambiguous definition. The Organization for Economic Cooperation and Development describes financial literacy as the combination of customers'/investors' understanding of financial products and concepts, their ability to appreciate financial risks and opportunities, and their confidence in their ability to make informed choices, know where to go for help, and take other effective actions to improve their financial well-being (Atkinson and Messy 2012). In the literature, the standard index measure used to examine objective financial literacy is that of Lusardi and Mitchell (2011). In our project we use the three principal questions developed by Lusardi and Mitchell (2011) and their widely used index to create an aggregated index variable. Given that the level of financial literacy may be correlated with the WTP for the different advising mechanisms, we control for that. The questions are as follows:

1. Suppose you had \$100 in a savings account and the interest rate was 2% per year. After 5 years, how much do you think you would have in the account if you left the money to grow?
 - A) More than \$102
 - B) Exactly \$102
 - C) Less than \$102
 - D) Do not know
 - E) Refuse to answer
2. Imagine that the interest rate on your savings account was 1% per year and inflation was 2% per year. After 1 year, how much would you be able to buy with the money in this account?
 - A) More than today
 - B) Exactly the same
 - C) Less than today
 - D) Do not know
 - E) Refuse to answer
3. Please tell me whether this statement is true or false. "Buying a single company's stock usually provides a safer return than a stock mutual fund."
 - A) True
 - B) False
 - C) Do not know
 - D) Refuse to answer

Notes: The questionnaire was developed by Lusardi and Mitchell (2011). It was translated and presented to the subjects in Hebrew.

Appendix E: Robustness Tests

Analysis by age group

Because there is no convention in the finance literature on the age group ranges in analyses of financial decision making, we provide an analysis of the RTA and WTP for four age groups (20–30 years, 31–40 years, 41–50 years, 51+ years) rather than the three used in our main analyses.

We observe that the results with these age groups are similar to those for three age groups. The 31- to 40-year-olds are less ready to adopt and less willing to pay for the algorithm alternative (0.50 and 6.08, respectively) compared to the human alternative (0.76 and 12.55, respectively). For the 41- to 50-year-olds we see that there is less of a WTP for alternatives that include an algorithm compared to purely human advice (mean WTP = 11.00, 10.44, and 14.86 for the algorithm, algo-human, and human alternative, respectively). Unlike for the three age groups, we see that for the 41- to 50-year-olds, adding human assistance does not change the subjects' perceptions. Subjects do not significantly prefer to adopt the algo-human alternative (mean RTA = 0.50 and 0.59 for the algorithm and algo-human alternatives). See Table E1.

[Table E1]

Round 2 analysis

It is common practice in experimental economics studies to treat the first round in a multi-round experiment as a learning trial and to analyze only later rounds. Here we provide an analysis as if the learning process involved only one round and not two in Study 1. The analysis of Round 2 data shows that the main results hold (Table E2).

[Table E2]

Table E1. Sample divided into four age groups

Variable	Age group			
	20–30 years	31–40 years	41–50 years	51 years and above
Readiness to adopt				
Algorithm	0.63	0.50*	0.77	0.73
Algo-human	0.65	0.59	0.78	0.61
Human	0.57	0.76	0.77	0.61
Willingness to pay				
Algorithm	10.84	6.08**	11.00*	10.20
Algo-human	10.76	11.82	10.44*	10.52
Human	8.96	12.55	14.86	10.44

* $p < 0.10$. ** $p < 0.05$.

Table E2. Results of Study 1, Round 2

Variable	Age group			Gender	
	20–30 years	31–44 years	45 years and above	Male	Female
Readiness to adopt					
Algorithm	0.59	0.55**	0.66	0.68	0.51**
Algo-human	0.60	0.68	0.63	0.62	0.63
Human	0.55	0.73	0.63	0.75	0.50***
Willingness to pay					
Algorithm	9.87	8.43**	9.41	11.04	7.60*
Algo-human	9.35	9.80*	9.69	8.50	10.58
Human	9.10	13.84	11.29	11.70	10.10

* $p < 0.10$. ** $p < 0.05$. *** $p < 0.01$.